

Updates and Current Trends of Lymphoma Care & Diagnostics

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❖ Colorado State University



- ❖ Private Practice Rotating Internship
- ❖ University of Illinois – Medical Oncology Residency
- ❖ Southwest Veterinary Oncology – Radiation Oncology Residency



- ❖ Medical and Radiation Oncologist Private Practice in Arizona and Texas
- ❖ Industry – Aratana Therapeutics, QBiotics Group



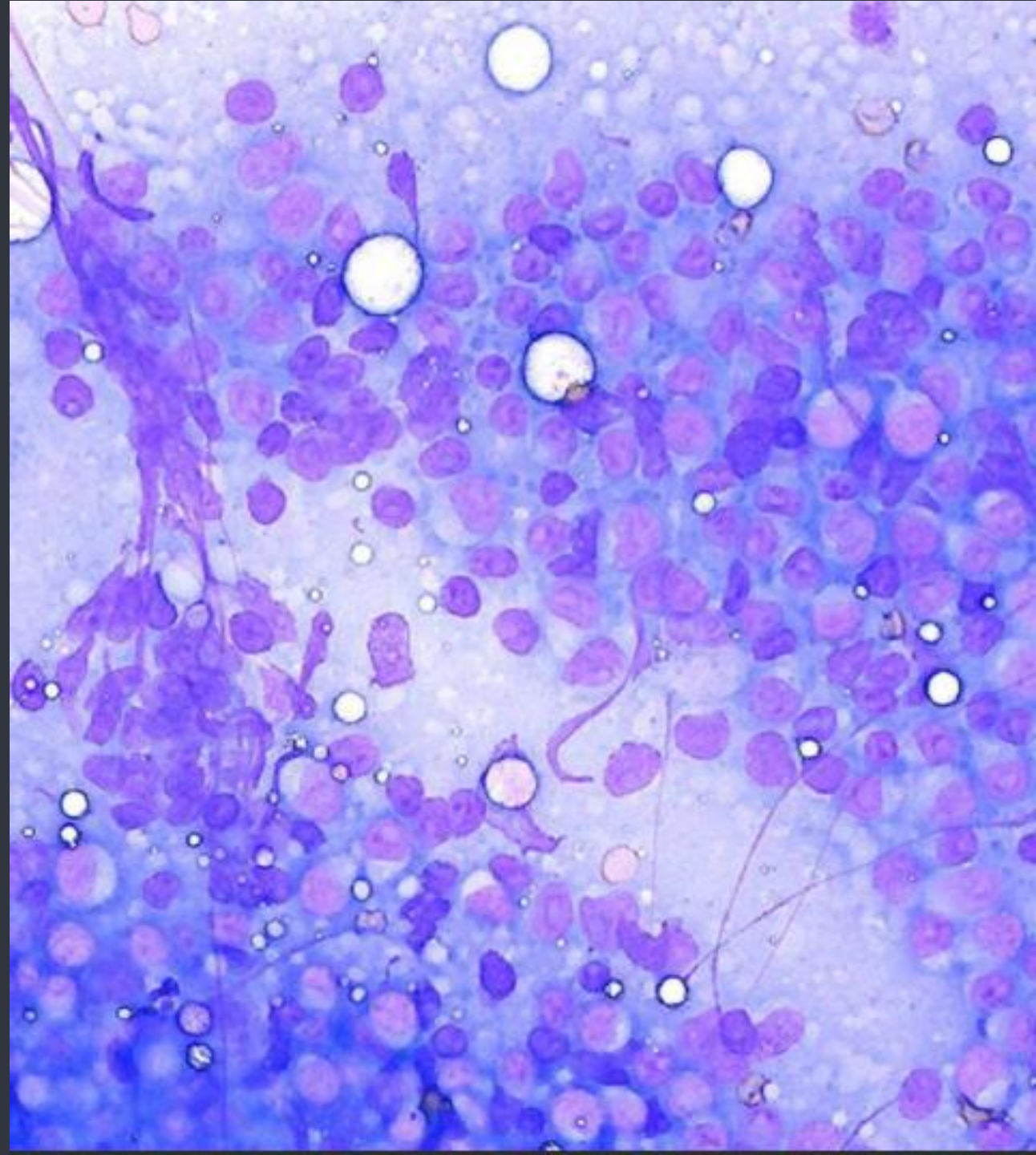
❖ IDEXX Specialist Support Veterinarian

Simple Approach to (nearly) all cancer

- + WHAT is it?
 - + Diagnose
- + WHERE is it?
 - + Staging tests
- + HOW do we treat it?
 - + Prognosis
 - + Treatment options
 - + Monitoring

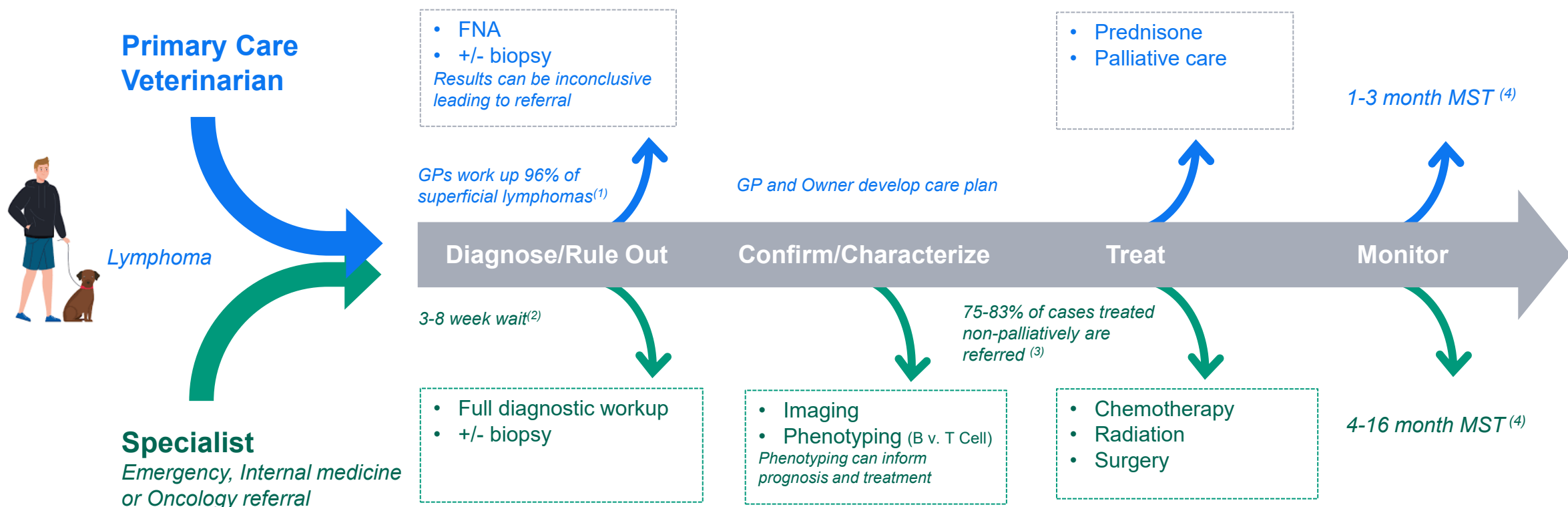


Foundations of lymphoma



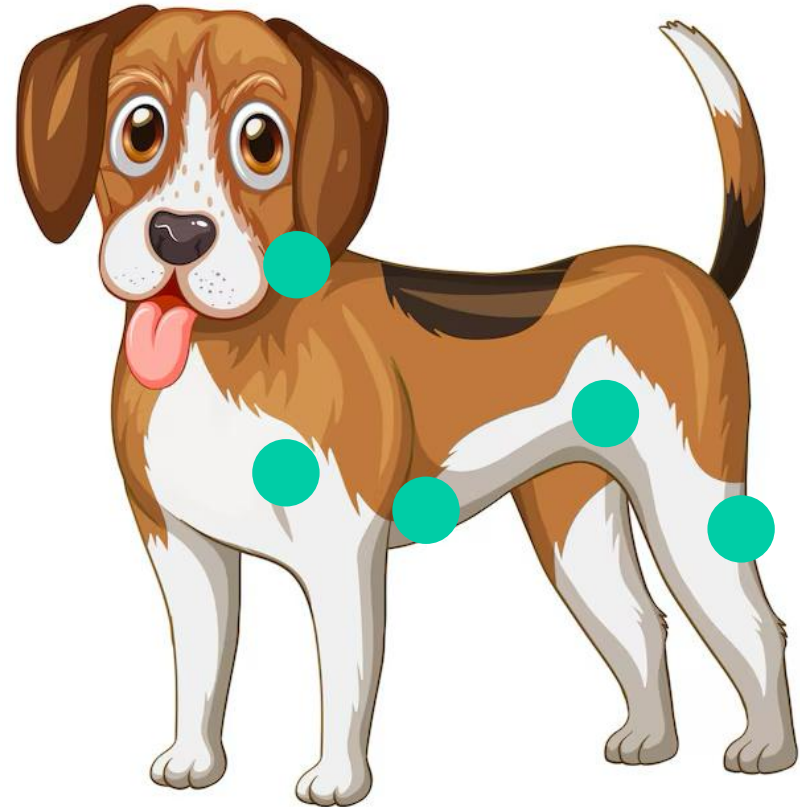
Shifting Focus: The Need for Earlier Detection in Cancer

In the case of Lymphoma, dogs present with enlarged lymph nodes at stage 3, 4 or even 5. **Pet owner conversations are focused on quality of life, extending limited time remaining**, and owner resource commitment for treatment options (finances and time). Most often **GPs will refer pet owners willing and able to treat** or you will manage palliatively with a focus on the best quality of life in the time remaining.



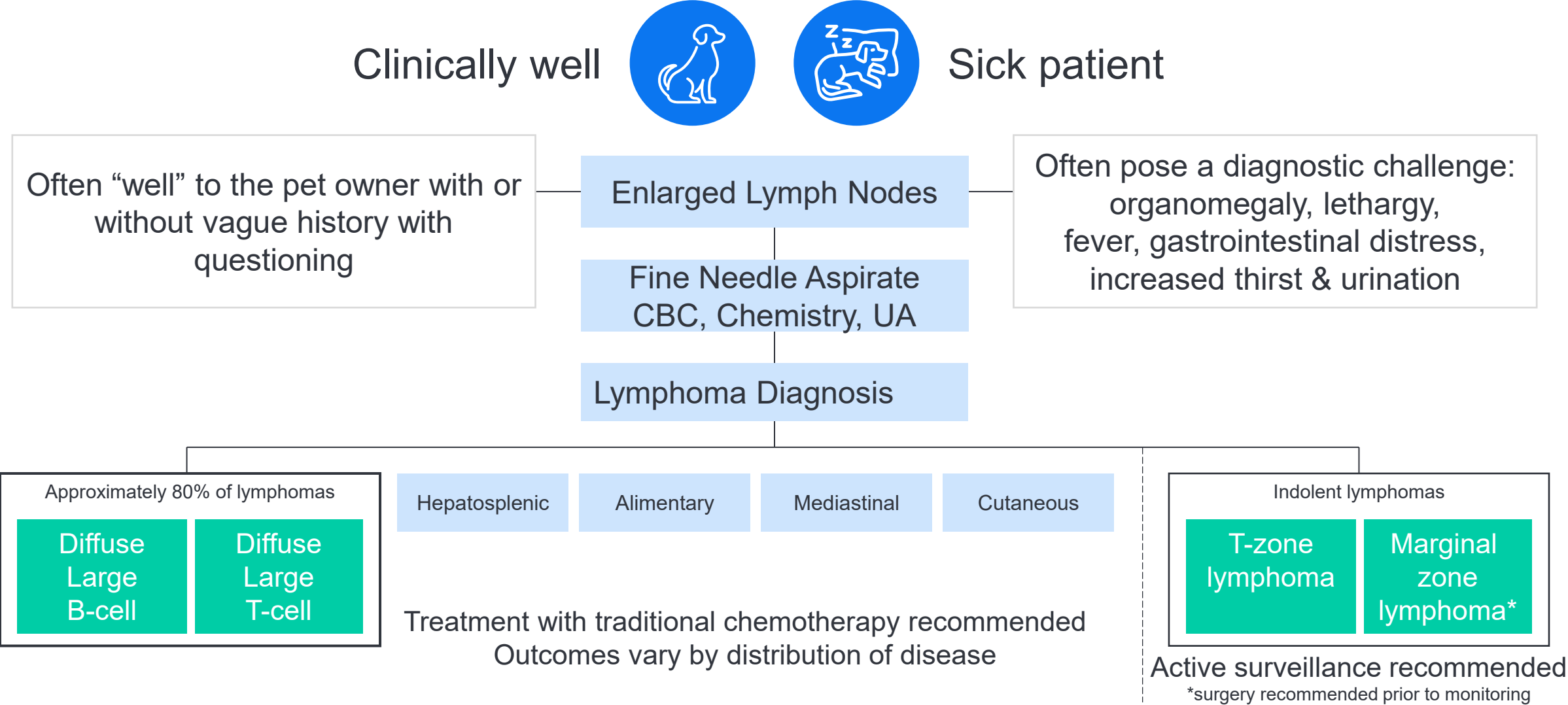
Lymphoma is a type of hematopoietic cancer, representing a malignant expansion of lymphocytes in the body.

Multicentric lymphoma is the most common form, having multiple centers of origin. The most common sites noted on physical examination are peripheral lymph nodes.



Locations of peripheral lymph nodes

The most common form is multicentric, but there are many subtypes which can complicate diagnosis



There are currently many diagnostics used to confirm lymphoma in dogs

Fine needle aspiration



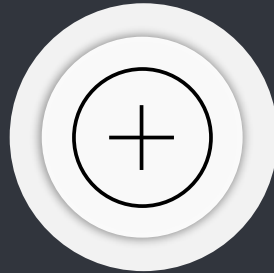
FNA for cytology –
most common
method of diagnosis
22 g needle on 6 mL
syringe

Biopsy



Used as a
secondary option
when FNA not
diagnostic
Additional
information about
tissue architecture

Immunohistochemistry



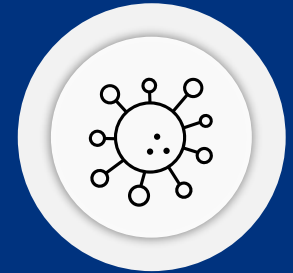
Performed as an
add-on test for
biopsy samples
Provides additional
prognostic
information
CD20, CD21,
CD79a, CD3

PARR



PCR for antigen
receptor
rearrangement
Performed on slides
Differentiates
inflammatory vs.
neoplastic
population
Provides additional
prognostic
information

Flow cytometry



Performed on live
cells
Provides additional
prognostic
information

Fine needle aspiration is most commonly the first diagnostic used to diagnose canine lymphoma

Fine needle aspiration



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Fine Needle Aspiration Clinical Questions

1

Is your sample diagnostic?

- + *What were you expecting?*
- + *Is that what you see?*

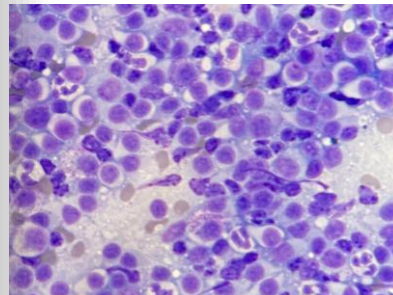
Next steps

If your sample is not diagnostic, consider reaspiration or biopsy

- + *What are some reasons the sample is not diagnostic?*

2

Is your sample consistent with lymphoma?



Consider immuno-phenotyping

What type of lymphoma does your patient have?

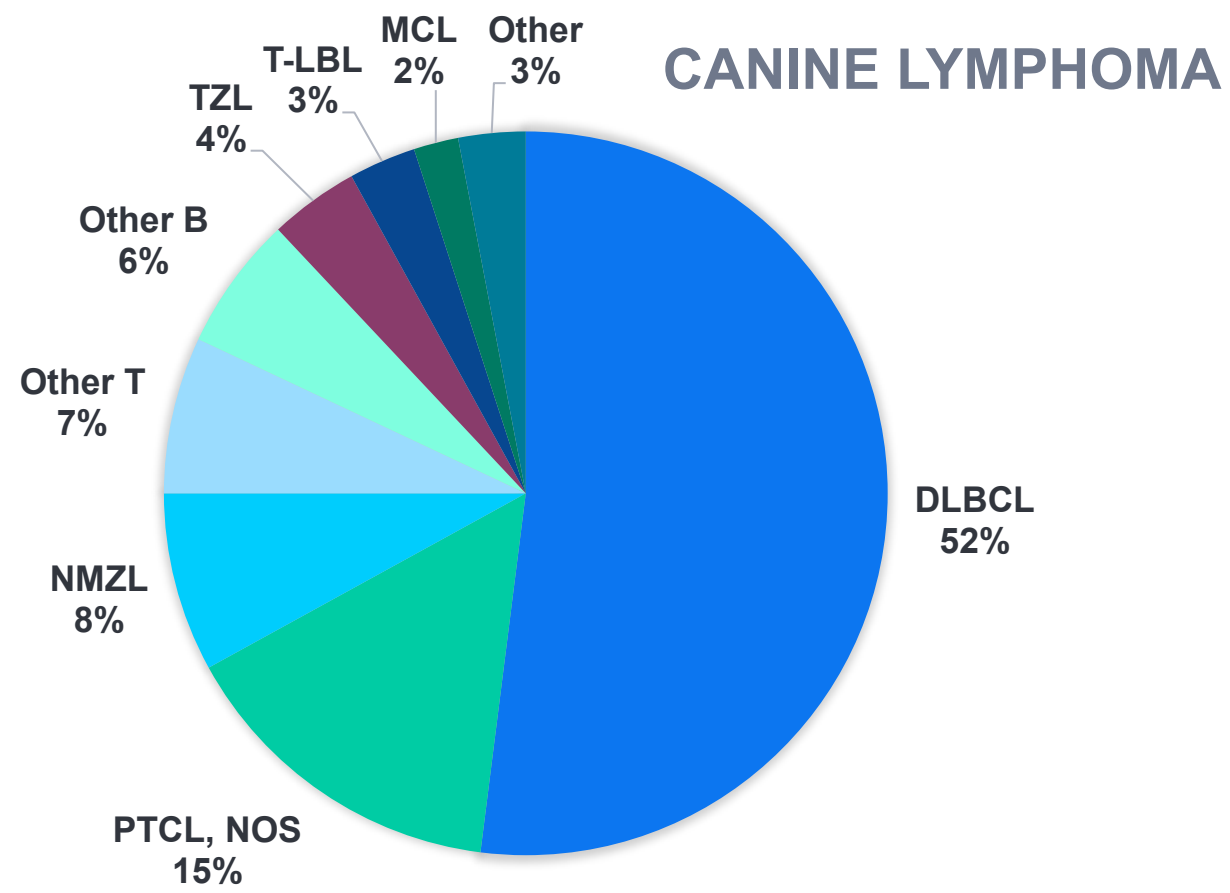
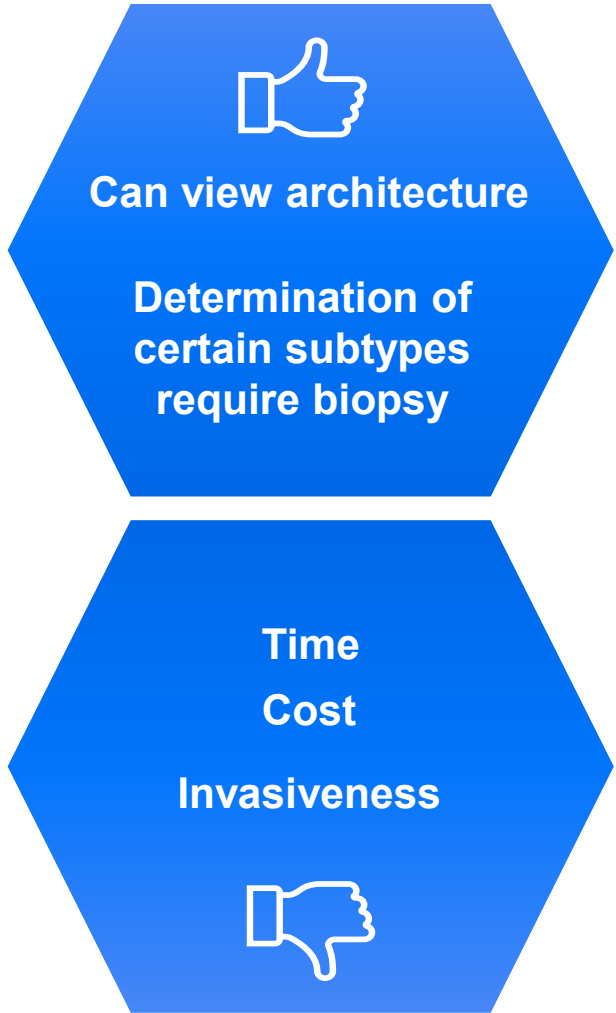
- + *What kind of sample do you have?*

Consider staging

How disseminated is the disease?

- + *What are the stages of lymphoma?*
- + *How do they correspond to diagnostics?*

Biopsy with or without immunohistochemistry



DLBCL= diffuse large B-cell lymphoma; PTCL, NOS= peripheral T-cell lymphoma, not otherwise specified; NMZL= nodal marginal zone lymphoma; TZL= T-zone lymphoma, T-LBL= T-cell lymphoblastic lymphoma; MCL= mantle cell lymphoma;

Qualifiers of diagnosis can help describe behavior and extent of disease



Immunophenotype
Type of lymphocyte
involved

Stage
How diffuse disease
is in the body

Substage
Clinical status of the
patient

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Why does immunophenotype matter?



Prognosis



Treatment



Some cases require additional characterization through biopsy with or without additional immunohistochemistry

Fine needle aspiration



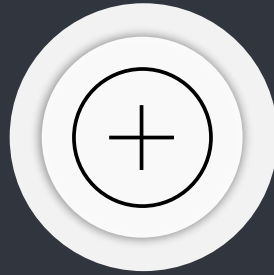
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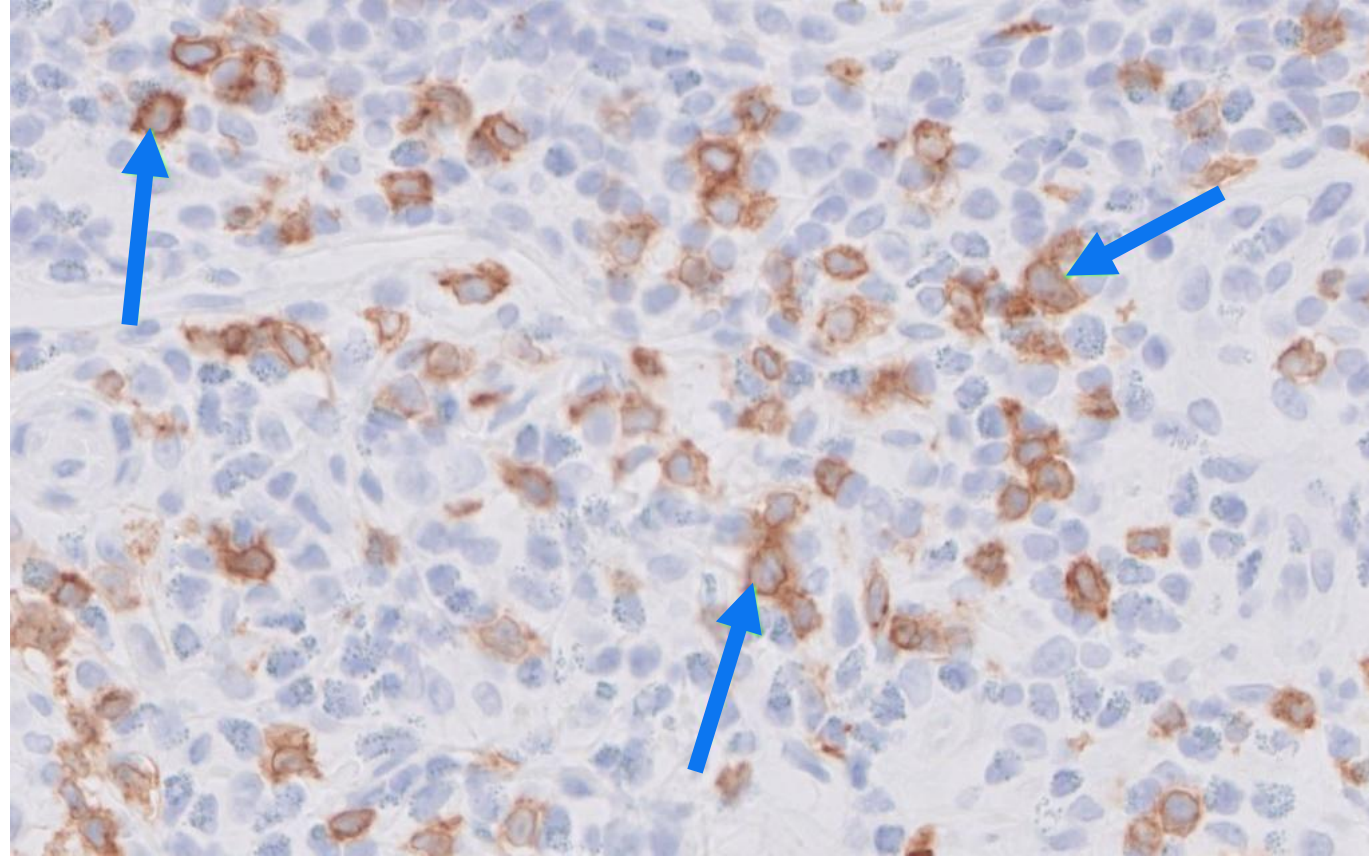
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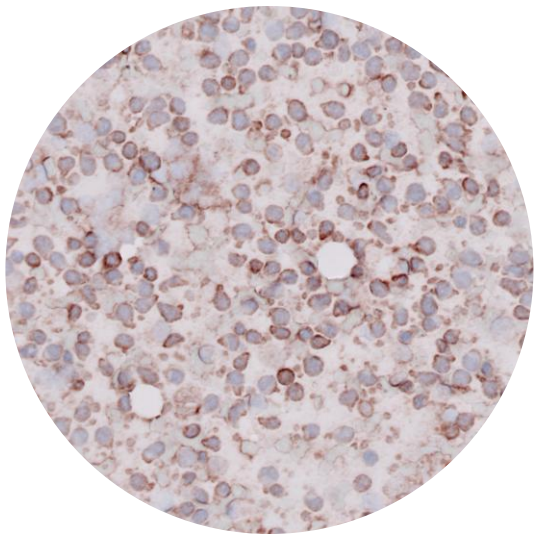
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Characterization of a T-cell lymphoma using immunohistochemistry



CD3 positivity (blue arrows) consistent with T-cell lymphoma

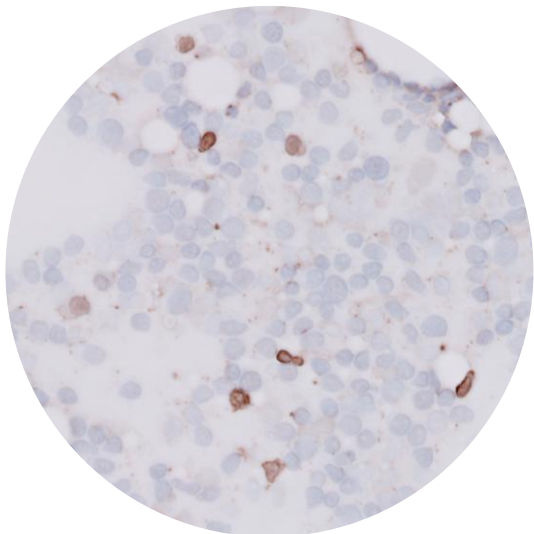
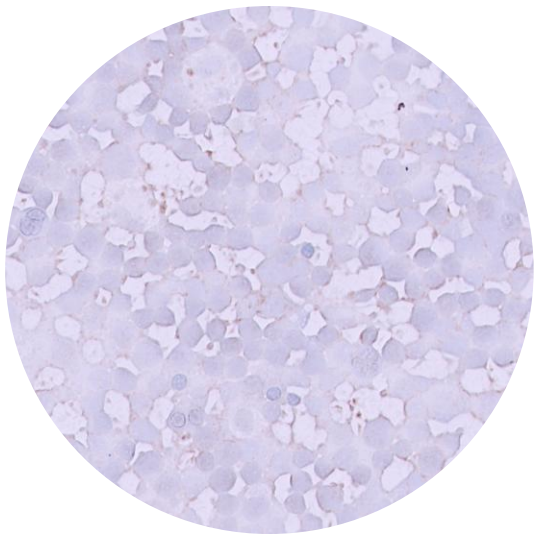
Characterization of lymphoma using immunocytochemistry



Positive

CD20 = marker
for **B-cells**

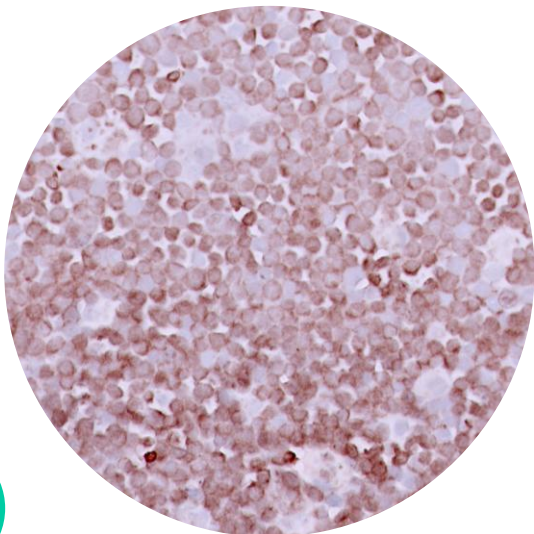
Negative



Negative

CD3 = marker
for **T-cells**

Positive



B-cell
lymphoma

T-cell
lymphoma

Additional characterization can be obtained with PARR and Flow cytometry

Fine needle aspiration



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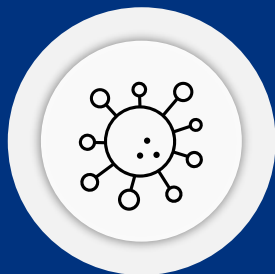
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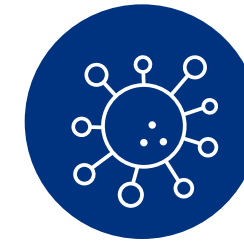
What are the differences between these diagnostics?



PARR

Does this patient actually have lymphoma?

- + Can be run on lower cellularity samples
- + Requires no special handling
- + TAT 10-14 business days
- + Specificity 89–94%^{1,2}



Flow Cytometry

What type of lymphoma does this patient have?

- + Gives the most comprehensive information about immunophenotype
- + Requires special handling
- + TAT 3-5 business days
- + Specificity 95–100%³

(1) CSU Clinical Hematopathology Laboratory, 2022

(2) Ehrhart EJ, Wong S, Richter K, Zismann V, Grimes C, Hendricks W, Khanna C. Polymerase chain reaction for antigen receptor rearrangement: Benchmarking performance of a lymphoid clonality assay in diverse canine sample types. J Vet Intern Med. 2019 May;33(3):1392-1402. doi: 10.1111/jvim.15485. Epub 2019 Apr 2. PMID: 30939225; PMCID: PMC6524097.

(3) Thalheim L, Williams LE, Borst LB, Fogle JE, Suter SE. Lymphoma immunophenotype of dogs determined by immunohistochemistry, flow cytometry, and polymerase chain reaction for antigen receptor rearrangements. J Vet Intern Med. 2013 Nov-Dec;27(6):1509-16. doi: 10.1111/jvim.12185. Epub 2013 Sep 20. PMID: 24112291.

There are multiple ways to obtain information about immunophenotype



Immuno- histochemistry (IHC)

- + Performed on biopsy tissue
- + Gold standard of phenotyping in canine lymphoma
- + Allows determination of subtype by architecture



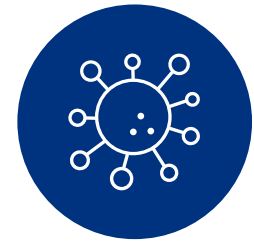
Immuno- cytochemistry (ICC)

- + Performed on cytology slides
- + Commonly offered on confirmed or suspicious cytology by clinical pathologist
- + CD20 (B), CD3 (T)



PARR

- + Performed on cells from biopsy or slides
- + Determines clonality of cell population
- + Used more commonly to determine if a population is cancerous



Flow cytometry

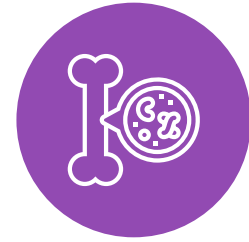
- + Performed on live lymphocytes suspended in saline and serum
- + Sorts lymphocytes by specific cell surface markers and size

Stages of lymphoma in dogs

Stage	Locations of disease	Method of Detection
1	Single lymph node	Physical examination/cytology
2	Multiple lymph nodes confined to one side of diaphragm	Physical examination/cytology
3	Diffusely enlarged lymph nodes	Physical examination/cytology
4	Liver and/or spleen involvement	Abdominal ultrasound
5	Bone marrow, ocular, CNS, lung, skin*	Chest radiographs, bone marrow aspirate

*Cutaneous lymphoma is described differently and not consistent with Stage 5 disease

Staging the lymphoma patient



Basic Lab Assessment

- + Complete blood count
- + Serum biochemistry
- + Infectious disease testing
- + Urinalysis

Chest radiographs

- + Three-view chest radiographs
- + Radiographs of other body compartments may be indicated based upon clinical signs

Abdominal ultrasound

- + Full abdominal ultrasound
- + May or may not include aspirates of organs

Bone marrow aspirate

- + Humerus or ilium
- + May or may not include a core biopsy

Abnormalities on minimum database

TEST	RESULT	
RBC	6.57	
Hematocrit	43.2	
Hemoglobin	15.1	
MCV	65.8	
MCH	23.0	
MCHC	35.0	
RDW	16.7	
% Reticulocytes	0.1	
Reticulocytes	8.5	L
Reticulocyte Hemoglobin	24.8	
WBC	3.15	L
% Neutrophils	66.6	
% Lymphocytes	25.7	
% Monocytes	5.4	
% Eosinophils	1.0	
% Basophils	1.3	
Neutrophils	2.10	L
Lymphocytes	0.81	L
Monocytes	0.17	
Eosinophils	0.03	L
Basophils	0.04	
Platelets	*43	L
PDW	- - -	
MPV	14.6	H

Bicytopenia—low neutrophils and low platelets

Clinical thought process:
Differentials include bone marrow involvement vs. secondary immune mediated process

- Suspect bone marrow involvement due to combination of neutropenia and thrombocytopenia

TEST	RESULT	
Glucose	103	
Creatinine	3.6	H
BUN	45	H
BUN: Creatinine Ratio	13	
Phosphorus	3.7	
Calcium	13.8	H
Sodium	153	
Potassium	4.0	
Na: K Ratio	38	
TEST	RESULT	
Chloride	116	
Total Protein	6.6	
Albumin	3.3	
Globulin	3.3	
Albumin: Globulin Ratio	1.0	
ALT	43	
ALP	74	
GGT	4	
Bilirubin - Total	0.3	
Cholesterol	171	
Amylase	497	L
Lipase	682	
Osmolality	313	

Hypercalcemia and azotemia

Clinical thought process:

Hypercalcemia likely secondary to malignancy (lymphoma and AGASACA* most common).

Azotemia could be pre-renal (dehydration), renal, or post-renal (obstruction). Likely renal secondary to hypercalcemia.

*Apocrine gland anal sac adenocarcinoma

What do chest radiographs tell us about canine lymphoma?

Mediastinal lymphoma

- + Mass effect on chest radiographs in cranial mediastinum
- + Fine needle aspiration could be considered
- + About 70% will have a concurrent hypercalcemia ⁽¹⁾
- + **About 97% of mediastinal lymphomas are T-cell in origin ⁽¹⁾**
- + **MST 183 days ⁽¹⁾**

Lung involvement

- + Unstructured interstitial pattern throughout lung fields
- + Bronchoalveolar lavage could be considered
- + Considered a feature of Stage 5 disease

Tracheobronchial lymph nodes

- + Enlarged lymph nodes caudal to carina at heart base
- + Difficult area to obtain diagnosis, consider bronchoscopy
- + L-asparaginase trial?



(1) Moore EL, Vernau W, Rebhun RB, Skorupski KA, Burton JH. Patient characteristics, prognostic factors and outcome of dogs with high-grade primary mediastinal lymphoma. Vet Comp Oncol. 2018 Mar;16(1):E45-E51. doi: 10.1111/vco.12331. Epub 2017 Jun 28. PMID: 28660709; PMCID: PMC5745293

Abdominal ultrasound- What does it tell us?

- + Lymphoma can involve ANY organ
- + Most common organs affected are liver and spleen
 - + Involvement of liver or spleen = Stage 4 disease
- + Gastrointestinal lymphoma
 - + Alteration or loss of wall layering
 - + Nonspecific findings?⁽¹⁾

Diagnostic Accuracy of Ultrasonography⁽²⁾

- + Detection of lymphomatous infiltration of liver
 - + Sensitivity 16.7%
 - + Specificity 91.0%
- + Detection of lymphomatous infiltration of spleen
 - + Sensitivity 73.1%
 - + Specificity 93.9%

Interesting Conclusions from Study

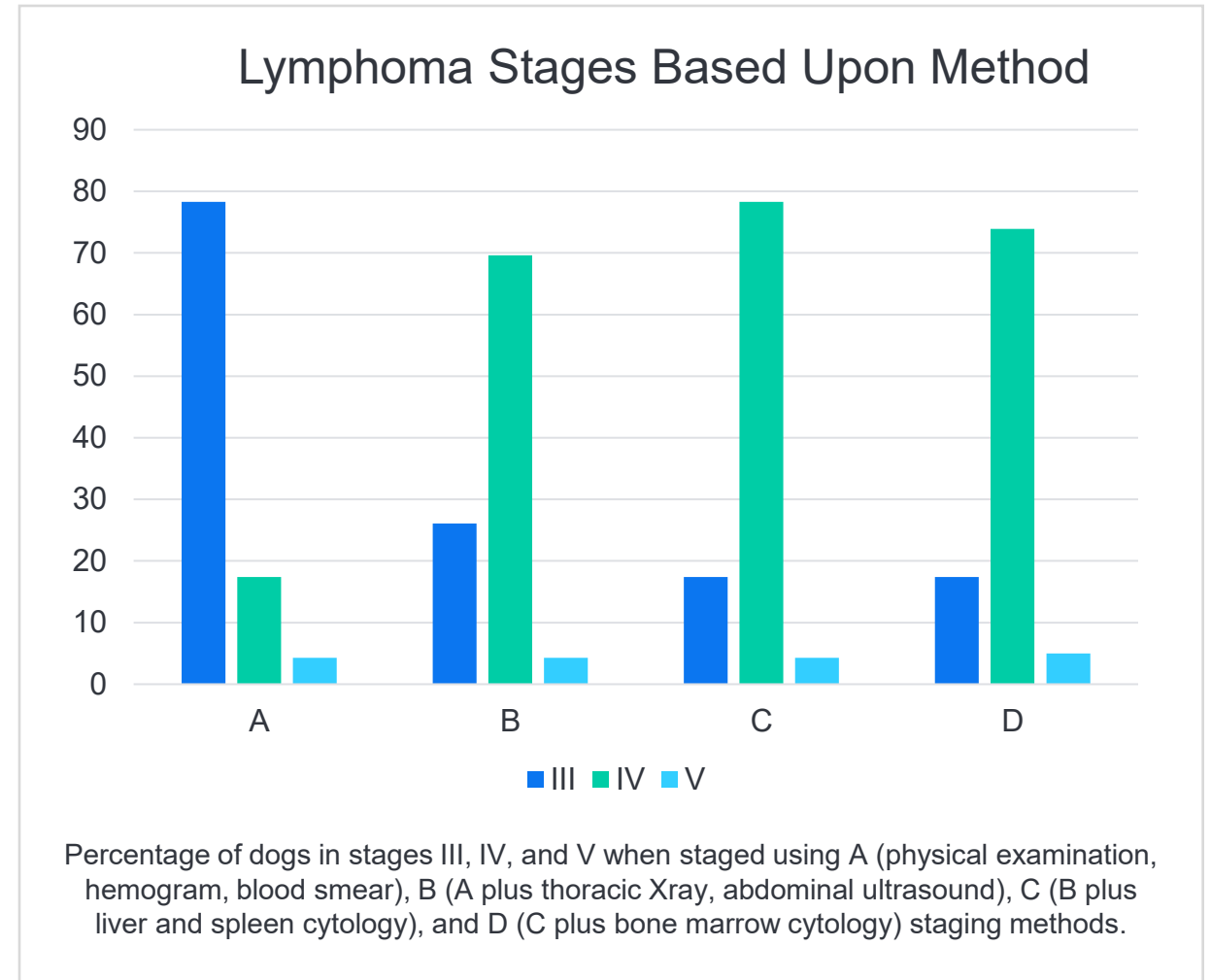
- + Ultrasonographically normal liver associated with no infiltration
- + Leopard-spotted spleen and splenomegaly associated with infiltration
- + Leopard-spotted spleen associated with B-cell

(1) Frances, M., Lane, A. E., & Lenard, Z. M. (2013). *Sonographic features of gastrointestinal lymphoma in 15 dogs*. *Journal of Small Animal Practice*, 54(9), 468–474. doi:10.1111/jsap.12117

(2) Sumping JC, Maddox TW, Killick D, Mortier JR. Diagnostic accuracy of ultrasonography to detect hepatic and splenic lymphomatous infiltration in dogs and cats. *J Small Anim Pract*. 2022 Feb;63(2):113-119. doi: 10.1111/jsap.13438. Epub 2021 Oct 7. PMID: 34617607.

Bone marrow aspirate—what information does it offer and is it necessary?

- + Not done often in practice. Why?
 - + Requires heavy sedation or anesthesia
 - + Invasiveness
 - + Costly to both the client and time of the clinician
- + Involvement = Stage 5 disease
 - + Stage migration
- + Other parameters serve as a surrogate to detect bone marrow involvement
 - + Thrombocytopenia and >10% lymphocytes
 - + Sensitivity 80%, specificity 89%



Pinto MT, Portillo I, Borrego J, Queiroga FL. Stage Migration in Canine Multicentric Lymphoma: Impact of Diagnostic Techniques on Assessing Disease Extent. In Vivo. 2024 May-Jun;38(3):1429-1435. doi: 10.21873/in vivo.13585. PMID: 38688606; PMCID: PMC11059893.

Substaging of dogs with lymphoma has been shown to affect prognosis



Substage a - Dog has no clinical signs upon presentation.



Substage b - Dog has clinical signs at presentation such as lethargy, gastrointestinal signs.

Substage	Number of dogs	Median Survival (Weeks)	Median Remission (Weeks)
a	31	69	44
b	24	28	18
		<i>P</i> = 0.001	<i>P</i> = 0.040

(1)



Inappetence
Vomiting
Diarrhea



Dyspnea



Changes in attitude
Weakness

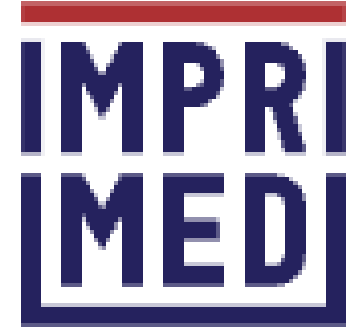
(2)

(1) Keller, E.T., MacEwen, E.G., Rosenthal, R.C., Helfand, S.C. and Fox, L.E. (1993), Evaluation of Prognostic Factors and Sequential Combination Chemotherapy With Doxorubicin for Canine Lymphoma. Journal of Veterinary Internal Medicine, 7: 289-295. <https://doi.org/10.1111/j.1939-1676.1993.tb01021.x>
(2) Barber LG, Weishaar KM. Criteria for designation of clinical substage in canine lymphoma: a survey of veterinary oncologists. Vet Comp Oncol. 2016 Aug;14 Suppl 1:32-9. doi: 10.1111/vco.12086. Epub 2014 Feb 23. PMID: 27508350.

Lymphoma treatments

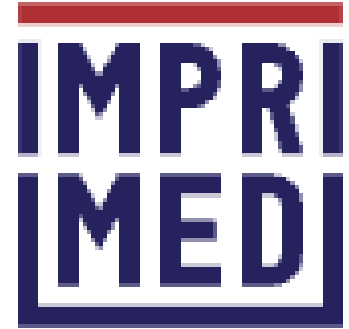


Imprimed



- + Chemotherapy Sensitivity Assays for Canine Lymphoma
 - + Personalized Prediction Profile
 - + CHOP protocol predictions
 - + Single agent drug response predictions
 - + Naïve and relapsed patients
 - + Samples needed:
 - + Lymphoma: FNA from swollen lymph node x 2 samples, 2 ml of EDTA blood
 - + Leukemia: 2 ml of EDTA blood
 - + Prior to treatment in naïve patients and relapse

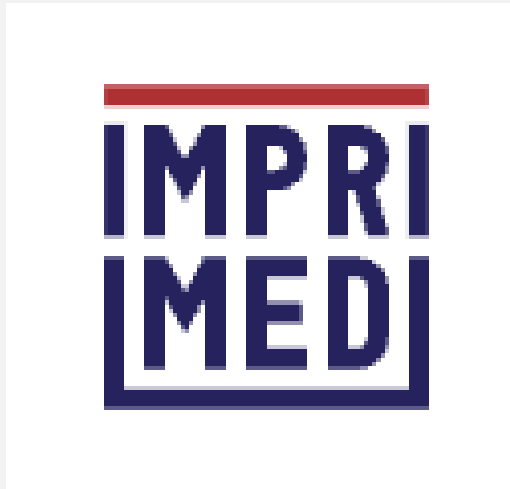
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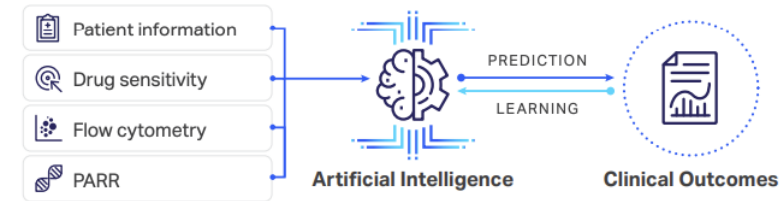
+ Chemotherapeutics Included:

- + L-Asparaginase
- + Mafosfamide (activated Cyclophosphamide)
- + Vincristine
- + Prednisolone (activated Prednisone)
- + Doxorubicin
- + Mitoxantrone
- + Rabacfosadine (Tanovea®)
- + Chlorambucil
- + Mechlorethamine (Mustargen)
- + Lomustine (CCNU)
- + Dexamethasone
- + Melphalan
- + Vinblastine

Personalized Prediction Profile Example

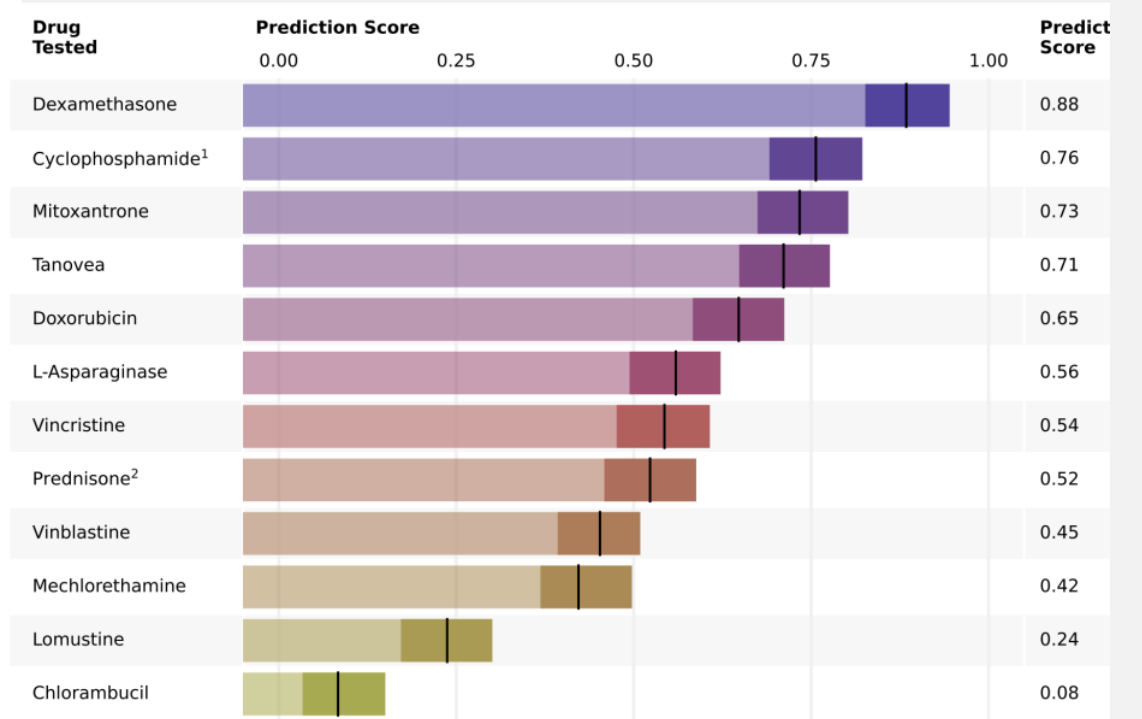


Clinical Outcome Predictions for Individual Drugs



ImpriMed's drug response prediction models are trained on real-world clinical responses with increasing amounts of information about patients. [Vet. Comp. Oncol. 2021; 19\(1\): 1-10](#), [Vet. Sci. 2021; 8\(12\): 301](#), [Front Oncol. 2024; 14:13041](#)

SINGLE DRUG RESPONSE PREDICTIONS



¹ Mafosfamide, a cyclophosphamide analog, was used to evaluate *in vitro* sensitivity of cyclophosphamide

² Prednisolone, an active form of prednisone, was used to evaluate *in vitro* sensitivity of prednisone

Treatments for lymphoma range from intensive to conservative

Multidrug protocol CHOP

Duration: 25 weeks

- + Doxo, CTX, Vinc, Pred
- + Sub in CCNU, Tanovea
- + One dose every week to every other week
- + Most dogs go into remission; <5% cured
- + MST - 14 mo (B-cell); 6–7 mo (T-cell)¹
- + \$3000–\$9000+ for course

Single agent protocol

Duration: 15 weeks

- + Doxorubicin, Tanovea, D+T alternating, CCNU
- + One dose every 3 weeks
- + Majority of dogs go into remission but less than multi-agent protocols
- + MST - 7–9 mo (B-cell); 4–5 mo (T-cell)¹
- + \$3000–\$6000+ for course

Oral options Laverdia

Duration: unknown

- + One dose twice weekly
- + More effective in T-cell than B-cell lymphoma
- + Paucity of literature on overall survival times in comparison to other protocols
- + \$160+ per month

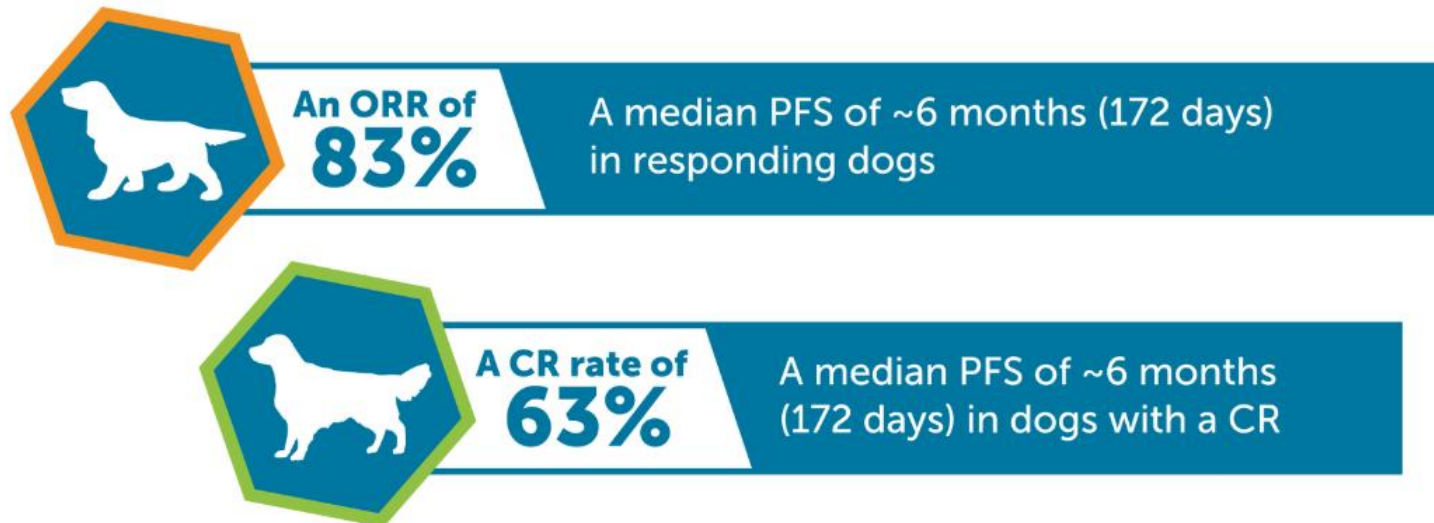
Palliative option Prednisone

Duration: unknown

- + One dose daily to twice daily
- + Some dogs will enter a brief partial or complete remission
- + MST 2–4 mo regardless of immunophenotype¹
- + \$30–\$200 per month

Tanovea (Rabacfosadine)

- + DNA polymerase inhibitor
- + IV infusion over 30 min every 3 weeks for 5 treatments
- + Side effects: vomiting, diarrhea, inappetence, lethargy
- + Unique toxicity – pulmonary fibrosis (West Highland White Terriers)



Bringing chemotherapy into the non-specialty clinic—Laverdia-CA1

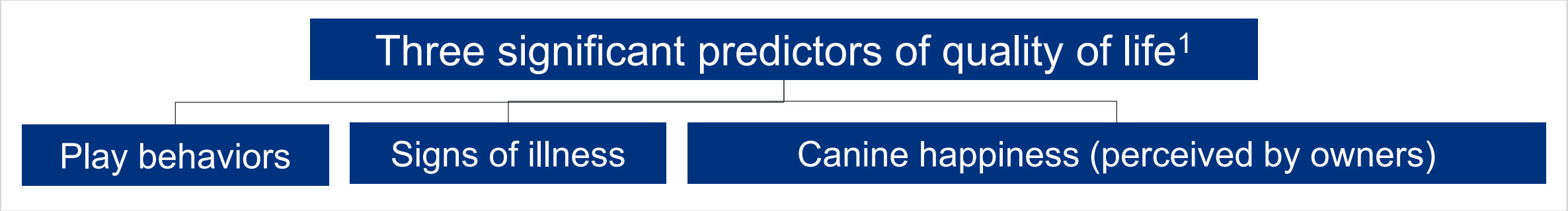
- + Conditionally approved for the treatment of canine lymphoma
- + Orally bioavailable
- + Selective inhibitor of nuclear export (SINE) of XPO1
 - + XPO1 overexpressed in many hematologic cancers
- + Included dogs: Naïve lymphoma or at first relapse
- + Overall response rate 37%
- + T-cell lymphoma overall response rate 71%
- + Well-tolerated



1.25 mg/kg administered orally twice per week

Sadowski AR, Gardner HL, Borgatti A, Wilson H, Vail DM, Lachowicz J, Manley C, Turner A, Klein MK, Waite A, Sahara A, London CA. Phase II study of the oral selective inhibitor of nuclear export (SINE) KPT-335 (verdinexor) in dogs with lymphoma. BMC Vet Res. 2018 Aug 24;14(1):250. doi: 10.1186/s12917-018-1587-9. PMID: 30143046; PMCID: PMC6109271.

Chemotherapy and cancer treatment do not negatively impact quality of life in the majority of patients



QOL is improved with chemotherapy ²	QOL does not negatively impact retreatment	Type of therapy does not influence QOL ³
+ 50% of dogs had improvement during chemotherapy protocol + Only 20% had a decline in QOL	+ 75% of clients would treat again with chemotherapy + Prolonged survival is important factor	+ Chemotherapy type (single vs. multi-agent) did not impact perception of QOL

(1) Iliopoulou MA, Kitchell BE, Yuzbasiyan-Gurkan V. Development of a survey instrument to assess health-related quality of life in small animal cancer patients treated with chemotherapy. J Am Vet Med Assoc. 2013 Jun 15;242(12):1679-87. doi: 10.2460/javma.242.12.1679. PMID: 23725431.

(2) Bergmann M, Sauter-Louis C, Hirschberger J. Lebensqualität und Lebenserwartung am malignen Lymphom erkrankter Hunde unter Chemotherapie. Eine Besitzerumfrage [Quality of life and life expectancy of dogs undergoing chemotherapy for malignant lymphoma. An owner survey]. Tierarztl Prax Ausg K Kleintiere Heimtiere. 2011;39(4):229-36. German. PMID: 22143660.

(3) Bianchi ML, Drudi D, Treggiari E, Catalucci C, Attorri V, Bonazzi I, Valenti P. Quality of life assessment in cancer patients receiving single-agent versus multidrug chemotherapy protocols. Open Vet J. 2021 Oct-Dec;11(4):755-763. doi: 10.5455/OVJ.2021.v11.i4.28. Epub 2021 Dec 5. PMID: 35070873; PMCID: PMC8770191.

Monitoring disease after treatment

Owner assessment at home

QOL, behavior, LN

Pet owner often serves as the “front-line” in early detection.

Periodic physical exam and LN assessment

Every 1-3 months

Lymph node enlargement is often the first sign of lymphoma coming out of remission.

Diagnostic tests

CBC/chemistry/UA

The timing and need for follow up lab diagnostics can depend on the patient, stage of disease and previous lab results. For example, patients that presented with hypercalcemia may need calcium level checked monthly.

Imaging – Abd Ultrasound, Thoracic Radiographs

Diagnostic imaging

Patients with stage IV or disease confined to the GI tract may need AUS and cranial mediastinal disease may need thoracic radiographs periodically.

IDEXX Cancer Dx™ testing



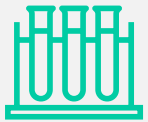
Veterinarians encounter multiple challenges in cancer diagnosis today



What is liquid biopsy?



- + Minimally invasive medical test that analyzes a sample of body fluid



- + Sample Types: Most commonly blood
- + Others: Urine, CSF, Intra-cavitary fluid



- + Detects signs of disease

June 2026

JAVMA

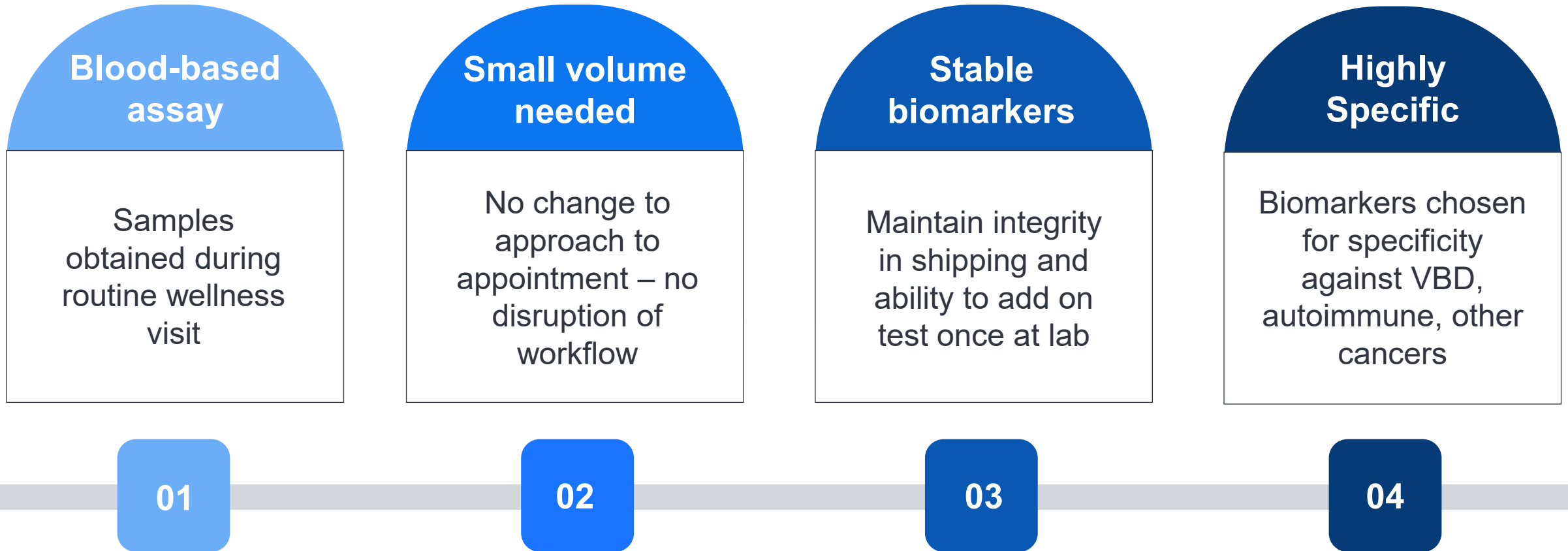
Currents in One Health

Leading at the intersection of
animal, human, and environmental health

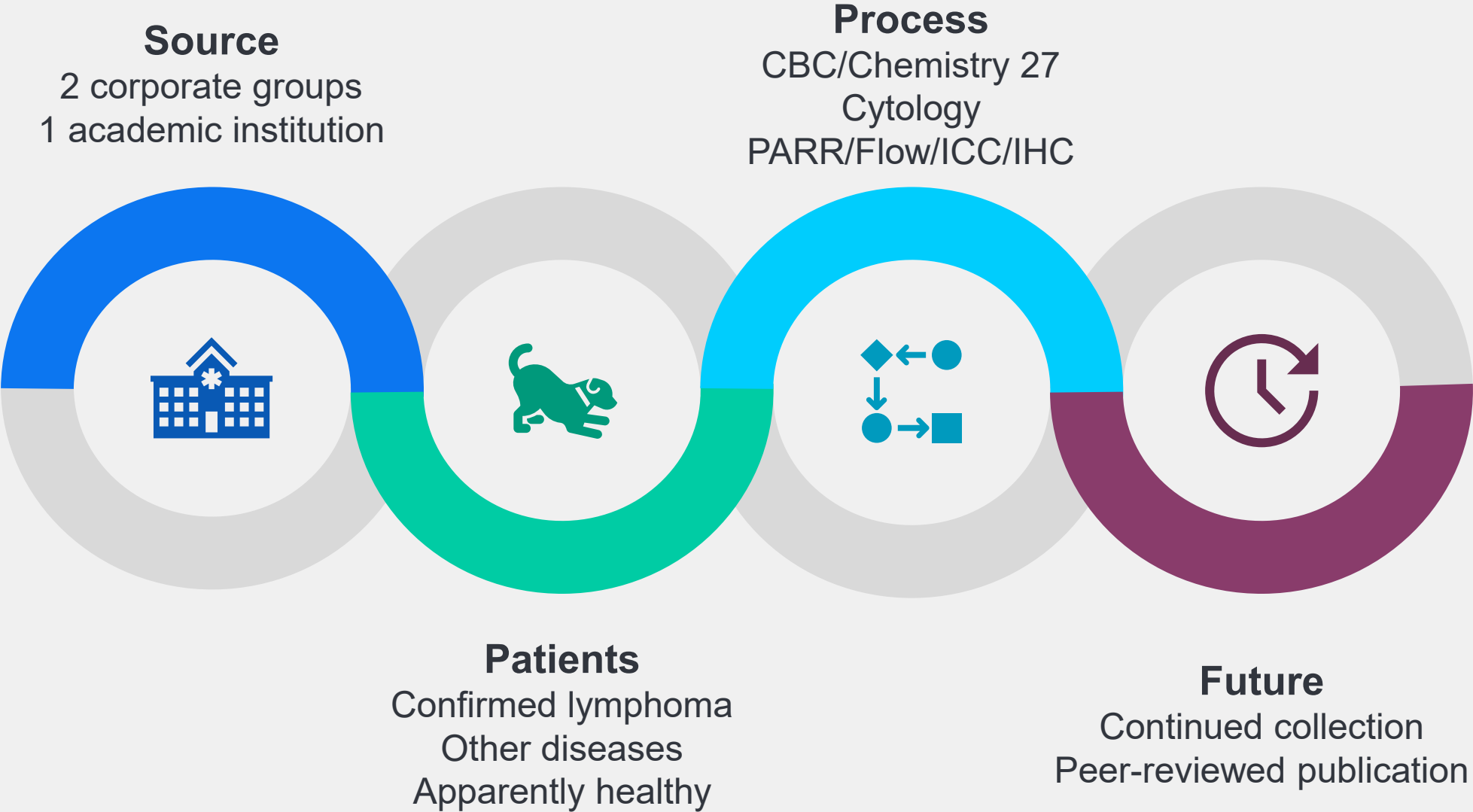


**Liquid biopsy technologies offer new insights and approaches
for canine cancer detection and management**

Requirements During Development of IDEXX Cancer Dx™



Validation and Verification of IDEXX Cancer Dx™



Population for lymphoma detection performance

105

Confirmed Lymphoma

- + Cytologic diagnosis + PARR
- + Cytologic diagnosis + Flow cytometry
- + Cytologic diagnosis + ICC
- + Histopathologic diagnosis + IHC
- + Naïve to treatment, off steroids and immunosuppressive > 1 month

73

Other Diseases

- + Inflammatory diseases
- + Other non-lymphoma cancers
- + Naïve to treatment with gross disease present

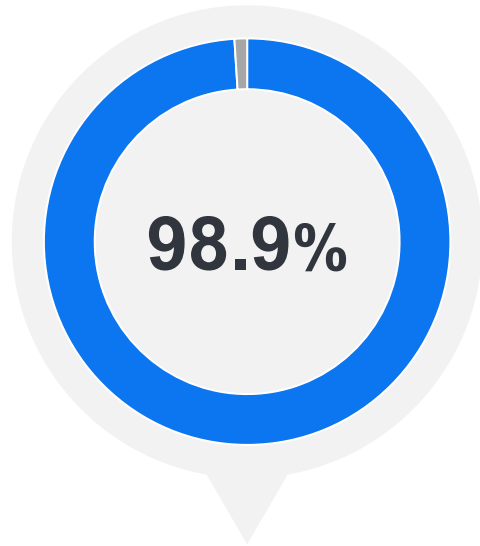
156

Apparently Healthy

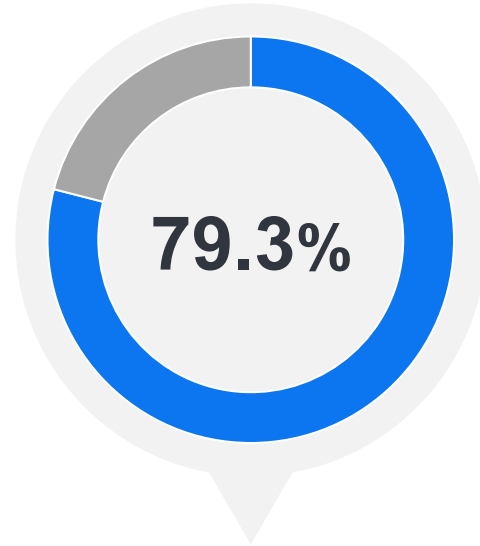
- + Normal physical examination
- + Normal CBC and full serum chemistry
- + Range of ages and breeds to reflect population seen in clinic

Additional dogs undergoing treatment for lymphoma were included in phenotype detection performance

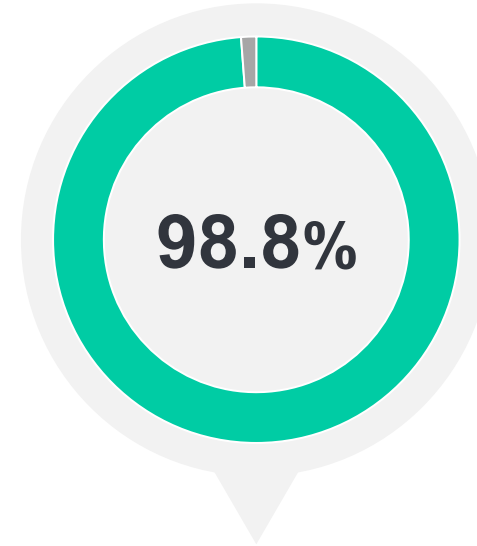
Verification and validation data



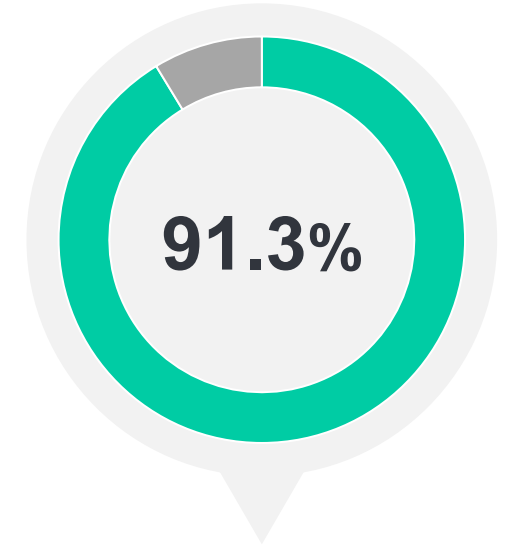
**LSA
Specificity**



**LSA
Sensitivity**



**T-cell
Specificity**



**B-cell
Specificity**

FP #1: Metastatic mast cell disease, PARR clonal for B-cell

FP #2: Splenic round cell tumor, suspected to be plasma cell in origin, MUM1 positive

56% IDEXX Cancer Dx™ results include a phenotype which is highly accurate at no additional cost to customer

Comparison to existing diagnostics

Where does IDEXX Cancer Dx™ fit into the diagnostic workup of a patient suspected to have lymphoma?

Metric	Cancer Dx	Cytology ⁽¹⁾	Lymphoma PCR (PARR) ⁽²⁻⁴⁾
Sensitivity	79%	92.6%	75% - 92%
Specificity	99%	89.4%	94% - 98.7%
Turnaround Time	2-3 days	1-5 days	10-14 days
Specimen	Serum/ Blood	Cells from lesion	Cells from lesion
Cost	\$	\$\$	\$\$\$

Where does IDEXX Cancer Dx™ fit into the available options for phenotyping of lymphoma?

Metric	Cancer Dx	Lymphoma PCR (PARR) ⁽⁵⁻⁶⁾	Flow cytometry ⁽⁶⁾
B-cell specificity	91.3%	67% - 89%	100%
T-cell specificity	98.8%	64% - 100%	98%
Specimen	Serum/ Blood	Cells from lesion	Cells from lesion
Special handling?	No	No	Yes
Cost	\$	\$\$\$	\$\$\$

1. Martini V, Marano G, Aresu L, Bonfanti U, Boracchi P, Caniatti M, Cian F, Gambini M, Marconato L, Masserdotti C, Nicoletti A, Riondato F, Roccabianca P, Stefanello D, Teske E, Comazzi S. Performance of lymph node cytopathology in diagnosis and characterization of lymphoma in dogs. J Vet Intern Med. 2022 Jan;36(1):204-214. doi: 10.1111/jvim.16326. Epub 2021 Nov 27. PMID: 34837263; PMCID: PMC8783335.

2. Avery A. Molecular diagnostics of hematologic malignancies. Top Companion Anim Med. 2009 Aug;24(3):144-50. doi: 10.1053/j.tcam.2009.03.005. PMID: 19732733

3. Frequently asked questions. Colorado State University Clinical Hematopathology Laboratory. Accessed 2/17/2025. <https://vetmedbiosci.colostate.edu/chl/faqs/>

4. Waugh EM, Gallagher A, Haining H, Johnston PEJ, Marchesi F, Jarrett RF, Morris JS. Optimisation and validation of a PCR for antigen receptor rearrangement (PARR) assay to detect clonality in canine lymphoid malignancies. Vet Immunol Immunopathol. 2016 Dec;182:115-124. doi: 10.1016/j.vetimm.2016.10.008. Epub 2016 Oct 19. PMID: 27863542; PMCID: PMC5119497.

5. Ehrhart EJ, Wong S, Richter K, Zismann V, Grimes C, Hendricks W, Khanna C. Polymerase chain reaction for antigen receptor rearrangement: Benchmarking performance of a lymphoid clonality assay in diverse canine sample types. J Vet Intern Med. 2019 May;33(3):1392-1402. doi: 10.1111/jvim.15485. Epub 2019 Apr 2. PMID: 30939225; PMCID: PMC6524097.

6. Thalheim L, Williams LE, Borst LB, Fogle JE, Suter SE. Lymphoma immunophenotype of dogs determined by immunohistochemistry, flow cytometry, and polymerase chain reaction for antigen receptor rearrangements. J Vet Intern Med. 2013 Nov-Dec;27(6):1509-16. doi: 10.1111/jvim.12185. Epub 2013 Sep 20. PMID: 24112291.

IDEXX Cancer Dx™ can be used on both well and sick patients



Screened based on population risk factors (no clinical signs)

- + Part of routine preventive care screening
- + Early studies show potential to detect signal weeks to months in advance of clinical manifestation of disease
- + At-risk dogs*

*At-risk dogs include all dogs ≥ 7 years old and high-risk breeds ≥ 4 years old.

Reference

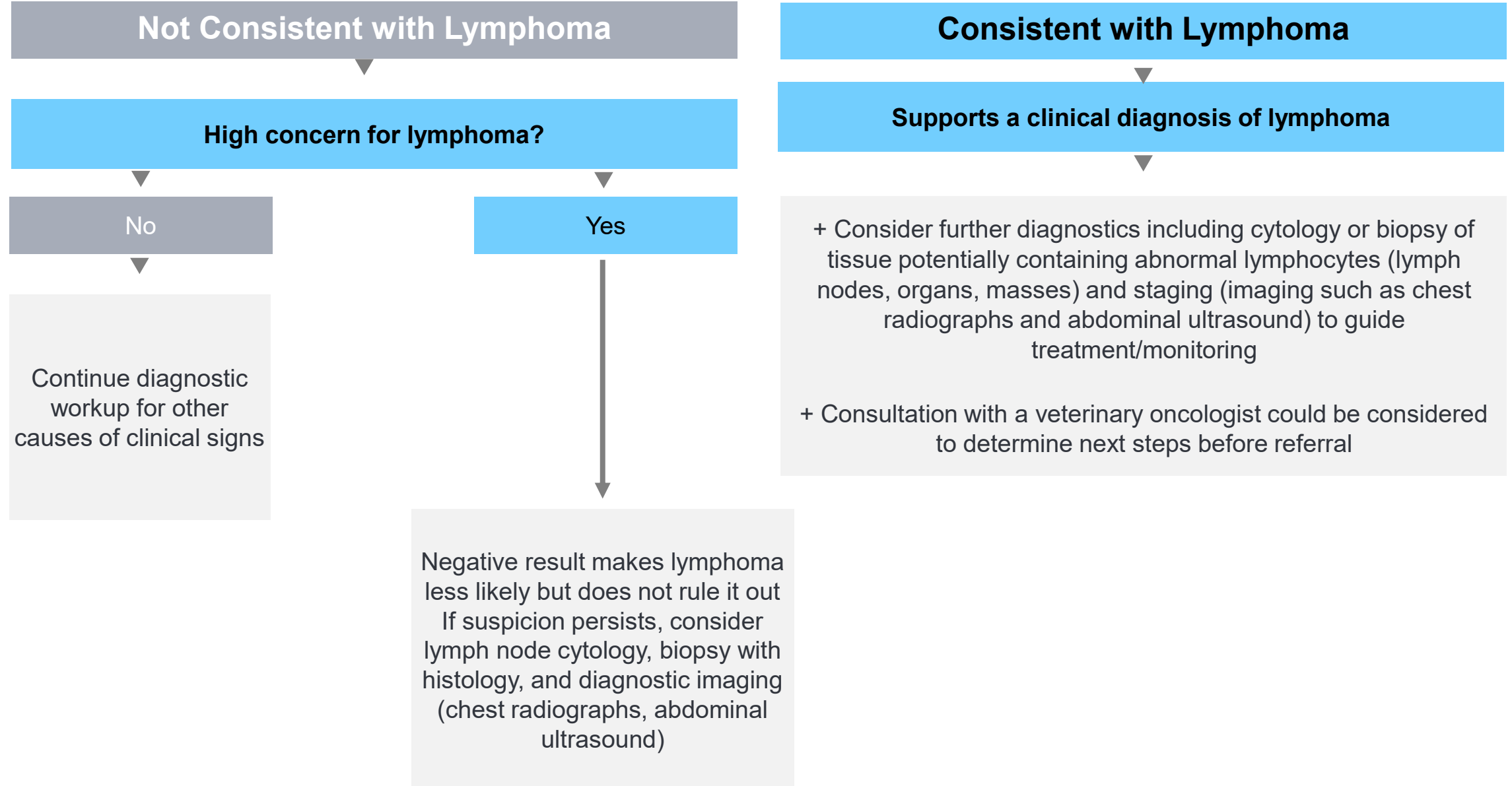
1. Data on file at IDEXX Laboratories, Inc. Westbrook, Maine USA: Data based on testing performed at IDEXX Reference Laboratories in North America between November 1, 2024, and December 6, 2024. *Analysis Report: IDEXX Cancer Dx Validation, 100282 [008_CancerDx-Validation-Report-2.Rmd]*.



Clinical signs suggestive of lymphoma

- + Part of diagnostic work-up
- + Accurate, efficient, affordable
- + Earlier detection¹
- + Increases access to cancer testing and care

Translating results into clinical decisions: Clinically ill dogs



Meet **Blaze**, a 5 y/o MN golden retriever



Initial visit – Day 1

Presenting complaint: Increased respiratory rate and effort

Chest radiographs: Pleural effusion

Thoracic and abdominal ultrasound: mediastinal mass and enlarged abdominal lymph nodes

Further diagnostics received – Day 2

Blood work with IDEXX Cancer Dx:

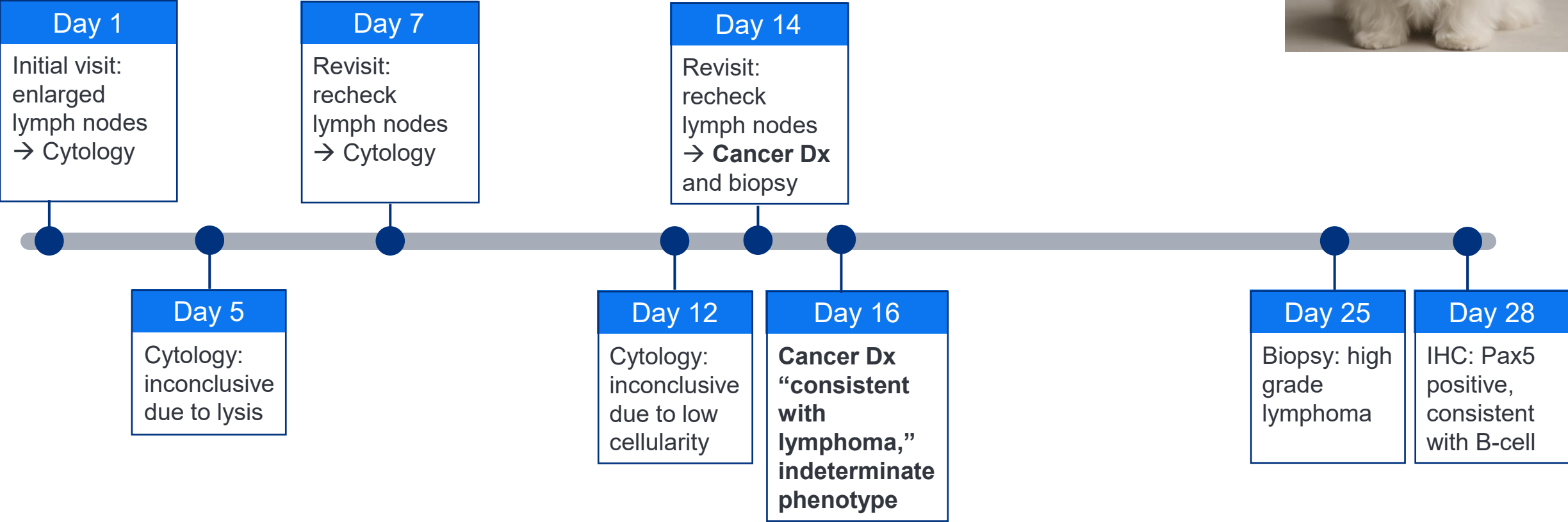
- CBC, Chemistry WNL, Lab4DX negative
- Cancer Dx consistent with lymphoma, T-cell phenotype

Further diagnostics received – Day 8

Flow cytometry at outside lab:

- T-cell lymphoma

Meet Evie, a 15 y/o FS Maltese



IDEXX Cancer Dx™ can be used on both well and sick patients



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Clinical signs suggestive of lymphoma

- + Part of diagnostic work-up
- + Accurate, efficient, affordable
- + Earlier detection¹
- + Increases access to cancer testing and care

Using IDEXX Cancer Dx™ in healthy patients

Why is cancer screening important?

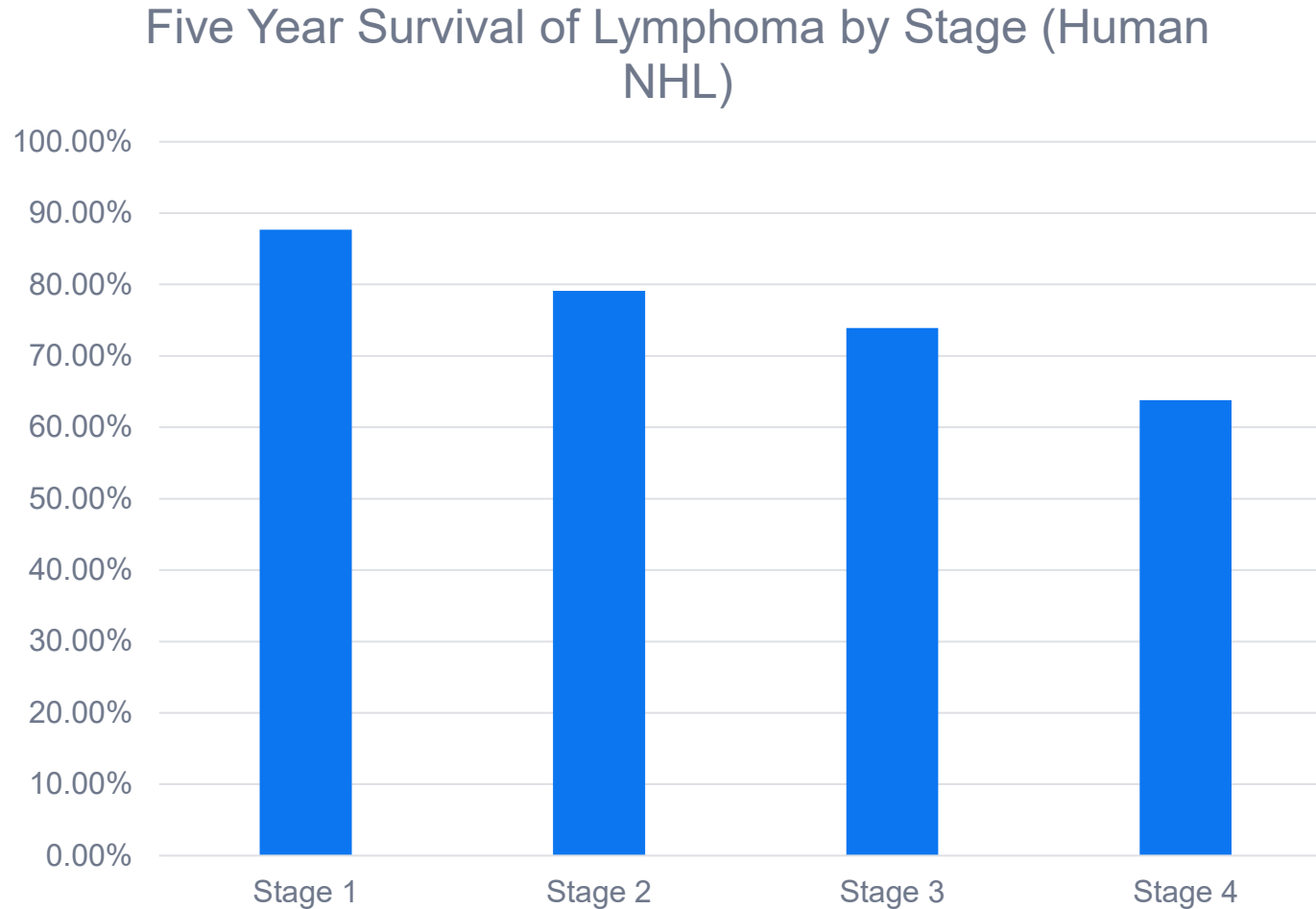
There is a shifting focus in oncology to early detection of cancers to provide more time for planning and a larger scope of opportunities to treat the disease. Diagnosing lymphoma before it can cause illness in a dog has a great potential to improve outcomes.

How does IDEXX Cancer Dx compare to other cancer screening strategies?

The IDEXX Cancer Dx test demonstrates performance that is comparable to, if not superior to, the current standards widely and consistently used in human cancer screening diagnostics today.⁽⁶⁻⁷⁾ IDEXX Cancer Dx also provides a lymphoma-specific result with additional characterization in a portion of cases at no additional cost.

Metric	IDEXX Cancer Dx	Mammography ⁽⁶⁾	Colonoscopy ⁽⁷⁾
Sensitivity	79%	65% - 94%	75% - 95%
Specificity	99%	50% - 98%	89% - 94%

Why Early Detection Matters



35%

stage 1 or 2 diagnoses in human non-Hodgkins lymphoma

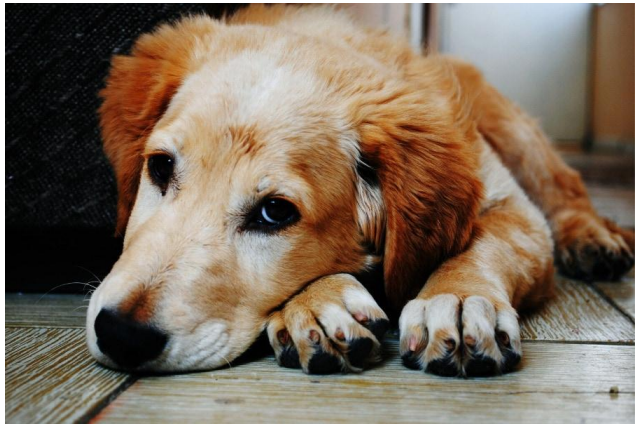
10% or less

stage 1 or 2 diagnoses in canine multicentric lymphoma

Substaging of dogs with lymphoma has been shown to affect prognosis



Substage a- Dog has no clinical signs upon presentation



Substage b- Dog has clinical signs at presentation such as lethargy, gastrointestinal signs

Substage	Number of dogs	Median Survival (Weeks)	Median Remission (Weeks)
a	31	69	44
b	24	28	18

P= 0.001 *P*= 0.040

(1)



Inappetence

Vomiting

Diarrhea



Dyspnea



Changes in attitude

Weakness

(2)

(1) Keller, E.T., MacEwen, E.G., Rosenthal, R.C., Helfand, S.C. and Fox, L.E. (1993), Evaluation of Prognostic Factors and Sequential Combination Chemotherapy With Doxorubicin for Canine Lymphoma. Journal of Veterinary Internal Medicine, 7: 289-295. <https://doi.org/10.1111/j.1939-1676.1993.tb01021.x>

(2) Barber LG, Weishaar KM. Criteria for designation of clinical substage in canine lymphoma: a survey of veterinary oncologists. Vet Comp Oncol. 2016 Aug;14 Suppl 1:32-9. doi: 10.1111/vco.12086. Epub 2014 Feb 23. PMID: 27508350.

The clinical blind spot



Majority of cancers in animals and humans are diagnosed late

This leads to the following:

- + Pressured decisions by pet owners
- + Decreased quality of life for patients
- + Decreased decisions to treat
- + Negatively impacted survival times

From gaps to clarity: a proactive approach

Early detection is the missing piece of the clinical puzzle in our cancer patients

Early cancer detection

- + Not routinely performed as it is in human medicine

Treatment

- + Therapy for cancer can include surgery, chemotherapy, radiation therapy, or palliative care



Confirmation

- + Diagnostics such as pathology and imaging confirm presence of disease

Monitoring

- + Examination and diagnostics are used to monitor remission status and recurrence

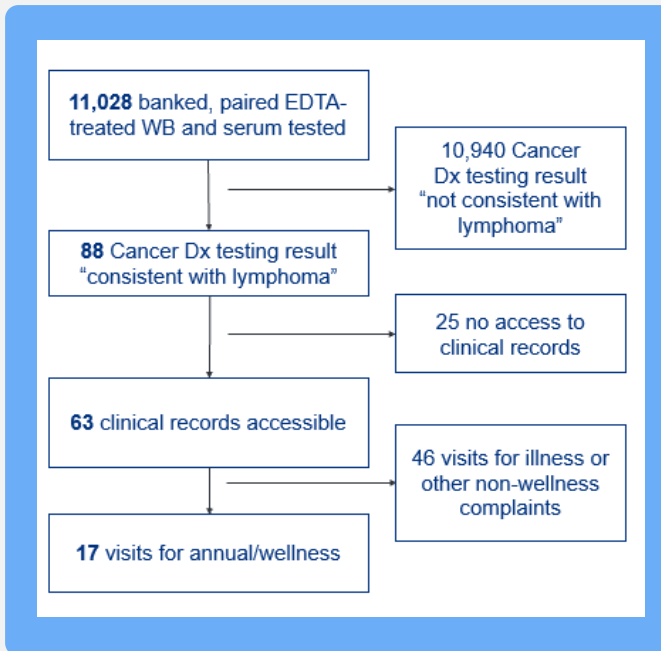
VCS 2025: POSTER ABSTRACT

Early Detection of Lymphoma by IDEXX Cancer Dx™ Testing in 7 Cases

Background

- + High specificity: 99%
- + Limited data in screening population
- + Case series describing patients with a Cancer Dx consistent with lymphoma on screening bloodwork

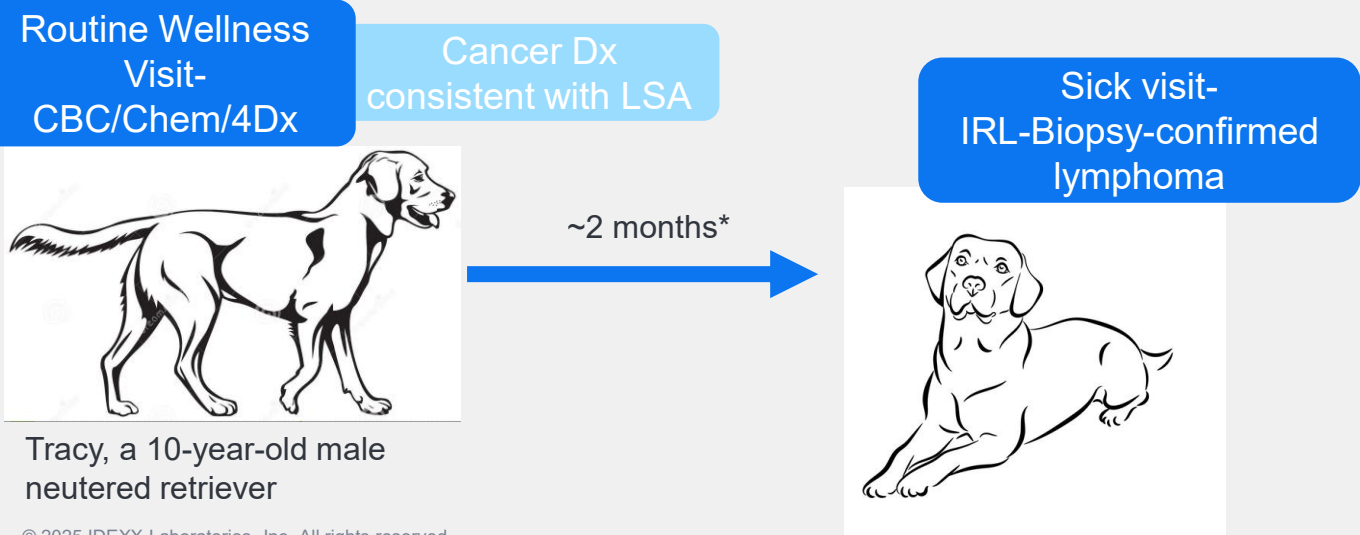
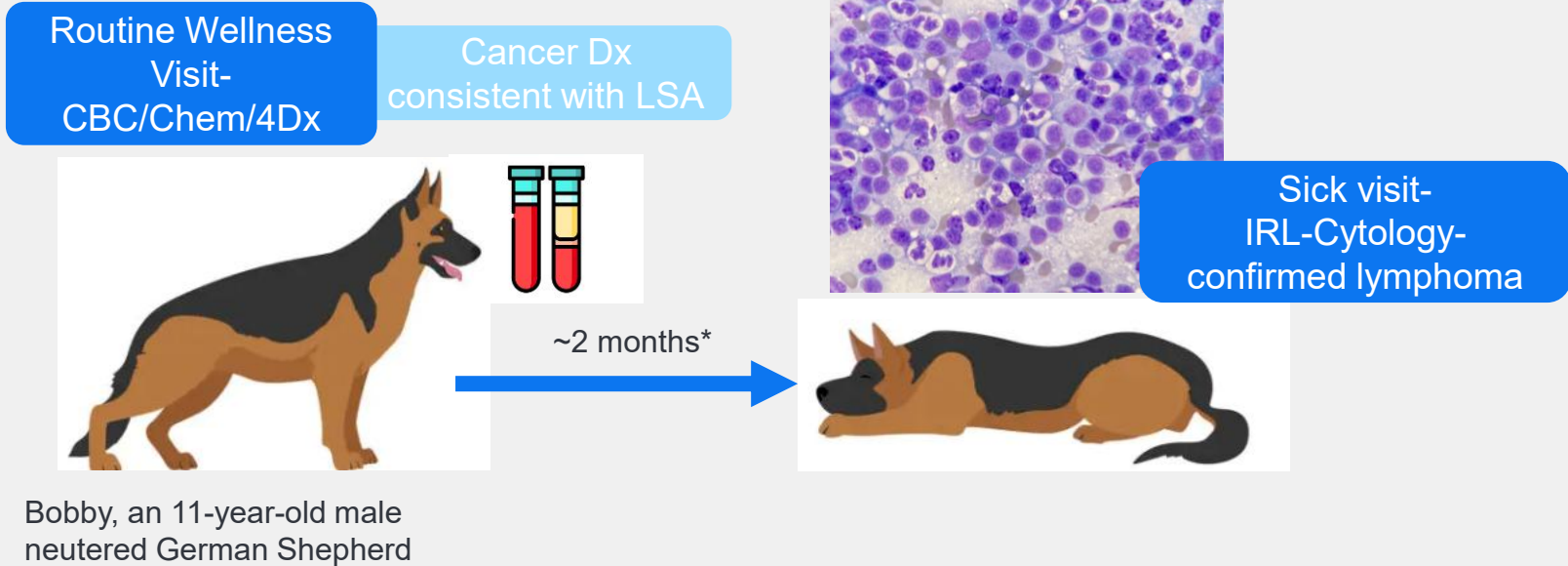
Methods



Results

- + Consistent with lymphoma: 88 dogs
- + Screening or vaccine visit: 17 dogs
- + Clinical diagnosis of LSA: 7/17 (41%)
- + Median time to diagnosis: 96 days

Early detection in Field Data Samples



June 19, 2024

Presenting complaint: Black gunk in mouth

Cancer Dx consistent with LSA



Burt, a 12-year-old male neutered Chihuahua

June 21, 2024

Dental prophylaxis and extractions performed

July 10, 2024

Presenting complaint: Declining with decreased appetite, generalized lymphadenopathy

Sick visit- Presumptive diagnosis of lymphoma- euthanasia

IDEXX 66

Early Detection of Lymphoma by IDEXX Cancer Dx™ Testing in 7 Cases

Discussion

- + IDEXX Cancer Dx testing has the potential to identify lymphoma before clinical suspicions in dogs.
- + Since this study used banked samples, Cancer Dx testing results did not influence the care that patients received.
- + Preliminary data suggests a Cancer Dx testing positive result may precede diagnosis of lymphoma by up to 6-8 months.
- + Close monitoring of apparently healthy patients with a positive Cancer Dx testing result for evidence of lymphoma is warranted.



VCS VSSO 2026: ORAL ABSTRACT

Initial Diagnostic Outcomes in Cancer Dx-Positive Patients Identified Through a Large-Scale Cancer Screening Study

Background

+ **SEARCHER study** aims to evaluate performance of IDEXX Cancer Dx™ testing as an early detection tool in dogs that are considered at risk based on age and/or breed

Methods

+ **Evaluate 5000 at risk patients** undergoing wellness examination and blood work

+ At risk defined as:

- ✓ ≥ 7 years old
- ✓ ≥ 4 years high-risk breeds

Results

+ **3,305 dogs** enrolled from April 1, 2025, to December 31, 2025

+ Consistent with lymphoma: **23 dogs**

+ Diagnostic work up: **17 dogs**

+ **5 dogs definitively diagnosed with lymphoid neoplasia**

Screening Efficacy of **At Risk Canines** undergoing **HEalthy Routine** examination



SEARCHER Study Design

Design

- ✓ **66** general practice clinics
- ✓ Patients undergoing **wellness exam** enrolled
- ✓ Blood work performed:
 - ✓ CBC with differential
 - ✓ Chemistry 25 or 27
 - ✓ IDEXX Cancer Dx™ testing
- ✓ Exclusion criteria:
 - ✓ Systemic treatment with immunosuppressives and/or steroids within 1 month of enrollment
 - ✓ Previous cancer diagnosis within 1 year OR any previous history of lymphoma

IDEXX Cancer Dx testing result: Consistent With Lymphoma

- + Diagnostic Work Up:
 - + Imaging, LN FNA/Cytology, Flow, PARR
- + Follow Up:
 - + 3 months
 - + 6 months

IDEXX Cancer Dx testing result: NOT Consistent With Lymphoma

- + 3 month Follow Up Survey
 - + Minimum 1500 patients
- + 6 month Follow Up Survey
 - + Minimum 1000 patients



SEARCHER STUDY

Positive Patient Work Ups

3,305 dogs enrolled from April 1, 2025,
to December 31, 2025

23 dogs had “Consistent with Lymphoma” result
17 dogs with further work up

Median time to work up: **20 days**
(range 4-71 days)

5 dogs definitively diagnosed
with lymphoid neoplasia (TP)

4 dogs presumptively
diagnosed with neoplasia

8 dogs with no evidence of
neoplasia at work up

Translating results into clinical decisions: Clinically **well** dogs

Not Consistent with Lymphoma

Result supports absence of detectable lymphoma;
evaluate patient at next planned wellness examination*

Consistent with Lymphoma

Evaluate full history, physical examination, and blood work

Further evaluation may be considered to confirm presence of disease, including the following:

- + Aspirates or biopsy of lymph nodes/organs
- + Diagnostic imaging (chest radiographs and abdominal ultrasound)
- + Flow cytometry of blood in cases of lymphocytosis

If lymphoma is not detected, perform recheck physical examination in 8-12 weeks
(sooner if new symptoms arise)

*American Animal Hospital Association (AAHA) recommends wellness examinations every 6 months in senior dogs and yearly in all other dogs.¹

Reference 1. Creevy KE, Grady J, Little SE, et al. 2019 AAHA Canine Life Stage Guidelines. J Am Anim Hosp Assoc. 2019;55(6):267–290. doi:10.5326/JAAHA-MS-6999

Early detection cases



Rose



Ruger



Rose's cancer story

Signalment: 11-year-old, spayed female Great Dane



November 19

Presenting complaint

- + Wellness examination and blood work

Physical examination

- + No significant abnormalities

Results available

- + CBC: no significant findings
- + Chemistry: no significant findings
- + IDEXX 4Dx™ Plus Test: negative

IDEXX Cancer Dx™ testing

- + Consistent with lymphoma

February 23

Presenting complaint

- + Hyporexia and weight loss for 2 weeks

Physical examination

- + Enlarged, firm peripheral lymph nodes

Results available

- + Fecal antigen testing: negative
- + Cytology: intermediate/large cell lymphoma

Ruger's cancer story

Signalment: 10-year-old, neutered male Aussie

Initial visit



Presenting complaint

- + Wellness examination

Physical examination

- + No abnormalities noted

Results available

- + CBC/chemistry: no significant findings

IDEXX Cancer Dx™ testing

- + Consistent with lymphoma, indeterminate phenotype

Workup discussed and pursued

Results available

- + Cytology of palpably normal node: **lymphoma**
- + Chest radiographs: normal
- + Abdominal ultrasound: mesenteric lymphadenopathy

IDEXX Cancer Dx results prompted a diagnostic workup leading to a ***diagnosis of lymphoma***, which was otherwise undetectable on physical examination and routine blood work

Pilot evaluation of a liquid biopsy test for longitudinal monitoring of disease status in dogs undergoing treatment for multicentric lymphoma

Study Objective

+ Investigate whether serial IDEXX Cancer Dx™ testing results correlate with clinical disease status in dogs receiving chemotherapy for multicentric lymphoma.

Methods

+ Prospective enrollment of **10 client-owned dogs** undergoing **chemotherapy**
+ Blood collection at first visit and subsequent chemotherapy or post-chemotherapy visits

Patient Characteristics

+ Median age: 6.5 years (range 2–14 years)
+ Stage: ≥ Stage 3
+ Lymphoma:

- + T cell: 1 patient
- + B cell: 9 patients

Reference

Nascimento, A., Peterson, S., Connell, D. *et al.* Pilot evaluation of a liquid biopsy test for longitudinal monitoring of disease status in dogs undergoing treatment for multicentric lymphoma. *BMC Vet Res* (2026). <https://doi.org/10.1186/s12917-026-05599-1>

Results: Remission outcomes



10 / 10 dogs

Achieved CR and converted to negative on IDEXX Cancer Dx™ testing



8 / 10 dogs

Achieved CR and converted to negative on IDEXX Cancer Dx testing **by week 5 of CHOP**



2 / 10 dogs

Achieved CR at week 2 and had delayed conversion to Cancer Dx testing negative **by weeks 6 and 7 of CHOP**



7 / 10 dogs

Maintained negative test results while remaining in CR on therapy

Results: Relapse correlation



- ✓ **7 / 10 dogs** had confirmed lymphoma relapse while enrolled in the study
 - ✓ **5 of these 7 dogs** were **positive** on **IDEXX Cancer Dx™** testing at the time of progressive disease
- ✓ **3 / 10 dogs** did not relapse during the study period and remained **negative** with **IDEXX Cancer Dx testing**
- ✓ This was an observational study, therefore treatment decisions were not influenced by IDEXX Cancer Dx testing results

Reference

Nascimento, A., Peterson, S., Connell, D. *et al.* Pilot evaluation of a liquid biopsy test for longitudinal monitoring of disease status in dogs undergoing treatment for multicentric lymphoma. *BMC Vet Res* (2026). <https://doi.org/10.1186/s12917-026-05599-1>

Pilot evaluation of a liquid biopsy test for longitudinal monitoring of disease status in dogs undergoing treatment for multicentric lymphoma

Discussion

- + In **all 10 dogs**, **IDEXX Cancer Dx™ testing** results converted to **negative** in correlation with clinical disease status with 2 out of 10 dogs having a delayed conversion.
- + In **5 out of 7 dogs** that had confirmed lymphoma relapse, **IDEXX Cancer Dx testing** results were **positive** at the time of documented relapse or disease progression.
- + These results highlight the potential of IDEXX Cancer Dx testing to monitor treatment response for multicentric lymphoma and support validation in larger cohorts.



Reference

Nascimento, A., Peterson, S., Connell, D. *et al.* Pilot evaluation of a liquid biopsy test for longitudinal monitoring of disease status in dogs undergoing treatment for multicentric lymphoma. *BMC Vet Res* (2026). <https://doi.org/10.1186/s12917-026-05599-1>

IDEXX

